

ABHINAV REDDY TERUPALLY

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SUMMARY

Data Science graduate focused on AI/ML, NLP, LLM systems, analytics, and responsible AI, with experience turning complex datasets and research workflows into reliable, explainable, production-minded applications.

TECHNICAL SKILLS

Programming: Python, SQL, R, C, JSON, REST APIs

Machine Learning & NLP: scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face, Transformers, Model Evaluation

GenAI & LLM Systems: LangChain, LangGraph, OpenAI API, Claude, Amazon Bedrock, RAG, Prompt Engineering, FAISS

Data Platforms & Analytics: PostgreSQL, MySQL, SQLite, Spark, pandas, NumPy, Power BI, DAX, Data Modeling, A/B Testing

MLOps, Monitoring & Version Control: FastAPI, Pydantic, Streamlit, Docker, DVC, MLflow, Prometheus, Grafana, Selenium, Git, GitHub, CI/CD

EDUCATION

University of Delaware

Aug 2024 – May 2026

M.S. in Data Science, GPA: 3.7/4.0

Newark, DE

Relevant Coursework: NLP, Machine Learning, Algorithms, Database Systems, Big Data Technologies

Sastra University

Jun 2019 – Jun 2023

B.Tech. in Computer Science and Engineering

India

EXPERIENCE

Research Assistant - University of Delaware

Sep 2025 – Present

Python, LangChain, NLP, Prompt Engineering, JSON, Model Evaluation

- Architected an end-to-end NLP/LLM pipeline that converts Stack Overflow posts into auditable sentence-level Entity-Aspect-Sentiment triples for software engineering research.
- Orchestrated JSON-first extraction workflows for entity detection, mention tracking, coreference resolution, aspect classification, sentiment labeling, schema validation, and traceable triple assembly.
- Benchmarked model outputs against human annotations, reaching 94.3% entity agreement, 96% aspect classification agreement, and 89.9% coreference decision accuracy.

Machine Learning Intern - Edureka

Aug 2024 – Aug 2024

Climate Change and Economy - Python, scikit-learn, Streamlit

India

- Developed a deployable climate-policy ML workflow across 141 countries by integrating climate, economic, agriculture, emissions, and carbon-pricing datasets into a reusable country-year pipeline.
- Implemented forecasting components, including an LSTM model that reduced next-year carbon pricing revenue prediction error by around 40%.

PROJECTS

Sanjaya AI - AI Advising Platform

Feb 2025 – May 2025

Python, FastAPI, Next.js, LangGraph, ChromaDB, RAG

- Engineered a full-stack AI advising platform that turns student profiles into prerequisite-safe semester roadmaps with citation-backed retrieval and advisor-style Q&A.
- Validated prerequisite closure, term-aware scheduling, credit limits, and Pydantic schema constraints outside LLM generation to keep recommendations deterministic and auditable.

E-Commerce Chatbot with Multi-Intent Classification

Aug 2025 – Sep 2025

LangChain, Selenium, SQLite, OpenAI, Streamlit

- Designed dual FAQ/SQL intent routing with semantic-router and all-MiniLM-L6-v2, achieving 82.5% routing accuracy, 0.80 Macro-F1, and 90% FAQ accuracy on paraphrase tests.
- Integrated OpenAI-powered NL-to-SQL search over a SQLite product catalog with clickable results, Selenium scraping, CSV-to-SQL ingestion, and fallback handling for unmatched or long-context queries.

Developer Experience in the AI Era: Power BI Analysis

Apr 2026 – Jun 2026

Power BI, DAX, Data Modeling, Survey Analytics

- Modeled 48K+ Stack Overflow Developer Survey 2025 responses into an interactive Power BI report covering AI adoption, sentiment, compensation, job satisfaction, experience groups, industries, and language/model preferences.

Credit Card Launch Impact Analysis

Oct 2025

Python, SQL, A/B Testing, Z-test

- Analyzed 500K+ transactions and 1,000 customer profiles to segment users for a new credit card launch, then verified a 100-customer 18-25 pilot with a Z-test showing a statistically significant 40% lift.

News Summarization Model Comparison

Mar 2024 – May 2025

Python, Fine-tuning, T5, GPT-2, ROUGE, BERTScore

- Fine-tuned T5-small and GPT-2 on Inshorts article-summary pairs with cleaning, tokenization, input formatting, GPU training, masked-label GPT-2 setup, and ROUGE/BERTScore evaluation; T5 achieved ROUGE-1 0.5365 and BERTScore 0.9061.